Daria Nogina

nogina.daria.ml@gmail.com | personal website | github | linkedin | google scholar

Education

Columbia University, PhD in Computer Science — **Sep 2025 – Present**
Advisor: Prof. Silvia Sellán; research in geometry processing and physical simulation — *New York, USA*

Lomonosov Moscow State University, BS + MS (with honors) — **2018 – 2025**
Bioengineering and Bioinformatics — *Moscow, Russia*

Selected Publications

Tube Maps: Fast SPH Boundary Handling in Tubular Coordinates — **SIGGRAPH 2026**

- **Daria Nogina**, Silvia Sellán. Geometry-based boundary handling method for smoothed particle hydrodynamics (SPH) that reduces boundary integrals to closed-form polynomials resulting in up to **100,000× lower boundary-handling cost** with comparable accuracy.

LegNet: a best-in-class deep learning model for short DNA regulatory regions — **Bioinformatics 2023**

- Dmitry Penzar*, **Daria Nogina***, et al. (*equal contribution). Developed a deep learning architecture for predicting gene expression from DNA sequences, achieving **first place** in the international machine learning hackathon.

Robust Curve Detection in Volumetric Medical Imaging via Attraction Field — **ShapeMI / MICCAI 2024**

- Farukh Yaushev, **Daria Nogina**, et al. Geometry-based detection of curves in volumetric medical images, applied to aorta centerlines.

Research & Industry Experience

Dandy — **Jul 2026 – Sep 2026**
ML Research Intern — constrained diffusion and 3D generation for dentistry — *New York, USA*

- Researched constrained diffusion generation, manuscript **submitted to ICLR 2027**.

IRA Labs — **Jul 2022 – Sep 2025**
ML Research Engineer — computer vision for medical imaging — *Moscow, Russia*

- Contributed to an aortic aneurysm detection system that surfaced missed diagnoses in archived scans, helping direct **at least five patients to emergency surgery**.
- Developed a novel aorta centerline detection method presented at a MICCAI workshop; supported more accurate detection and retrospective analyses.
- Designed and deployed a kidney-lesion detection service for CT scans; developed geometry-aware diaphragm localization improving related products by **up to 10%**.

EPFL — École polytechnique fédérale de Lausanne — **Jul 2024 – Sep 2024**
Summer@EPFL Research Intern; advisor: Prof. Maria Brbić — *Lausanne, Switzerland*

- Built a multimodal framework combining drug-molecule representations and microscopy images to predict cell morphology under drug perturbations.

VIGG RAS — Computational Systems Biology — **Jun 2019 – Jun 2023**
Researcher; advisor: Ivan Kulakovskiy — *Moscow, Russia*

- Studied regulatory sequences and translation, developing high-throughput analysis pipelines for ribosome-profiling and gene-expression data.

European Molecular Biology Laboratory (EMBL) — **Jun 2021 – Nov 2021**
Summer Research Intern; advisor: Judith Zaugg — *Heidelberg, Germany*

- Improved evaluation of cell-type-specific gene-expression prediction by identifying hyperparameter selection and data-splitting issues.

Selected Honors & Teaching

- **Honors:** DREAM 2022 competition 1st place; MSU department rank #1 (2019) and #2 (2023); Kaulen young-researcher best paper award; Summer@EPFL (1.6% acceptance rate).
- **Teaching:** Designed and taught a master's-level course in machine learning for medical imaging at Lomonosov MSU; mentoring of master's students at Columbia (see preprint *PointGrade: Geometric Priors for Grading MoonBoard Problems*).

Additional Publications & Research

- *A community effort to optimize sequence-based deep learning models of gene regulation.* **Nature Biotechnology**, 2024.
- *GRaNIE and GRaNPA*. **Molecular Systems Biology**, 2023; *Ribosomal leaky scanning through a translated uORF requires eIF4G2*. **Nucleic Acids Research**, 2022.
- Earlier work: comparative genomics of RNA editing in cephalopods (2019). Full bibliography: Google Scholar.